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(54) **BULK COOKING OIL DISTRIBUTION SYSTEM**

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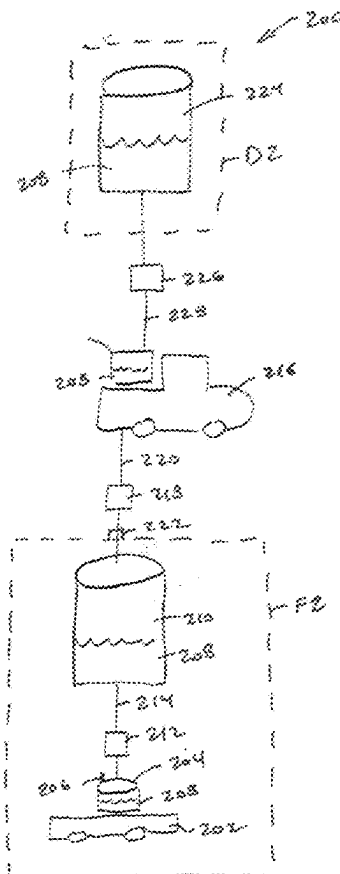
(57) **ABSTRACT**

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Related U.S. Application Data

(63) Continuation of application No. 18/376,743, filed on Oct. 4, 2023, now Pat. No. 12,330,547, which is a continuation of application No. 17/838,175, filed on Jun. 11, 2022, now Pat. No. 11,787,326, which is a continuation of application No. 17/103,594, filed on Nov. 24, 2020, now Pat. No. 11,364,836, which is a continuation of application No. 16/505,443, filed on Jul. 8, 2019, now abandoned, which is a continuation of application No. 15/212,550, filed on Jul. 18, 2016, now Pat. No. 10,343,584, which is a continuation of application No. 13/461,442, filed on May 1, 2012, now Pat. No. 9,392,907.

A portable cooking oil distribution device for filling remote fryers with fresh cooking oil. In other embodiments the device can also collect used cooking oil from the fryer. In one embodiment, a portable device includes a fresh oil tank, a three-way valve connected by piping to the fresh oil tank, a pump connected by piping to the three-way valve, and a nozzle connected by piping to the pump, the nozzle having an outlet tube connected to the piping between the pump and the nozzle. The device may also include a used oil tank, a second three-way valve connected by piping to the used oil tank, a second pump connected by piping to the three-way valve, and piping connecting the second pump to the nozzle, where the nozzle includes an inlet tube connected to the piping between the pump and the nozzle.



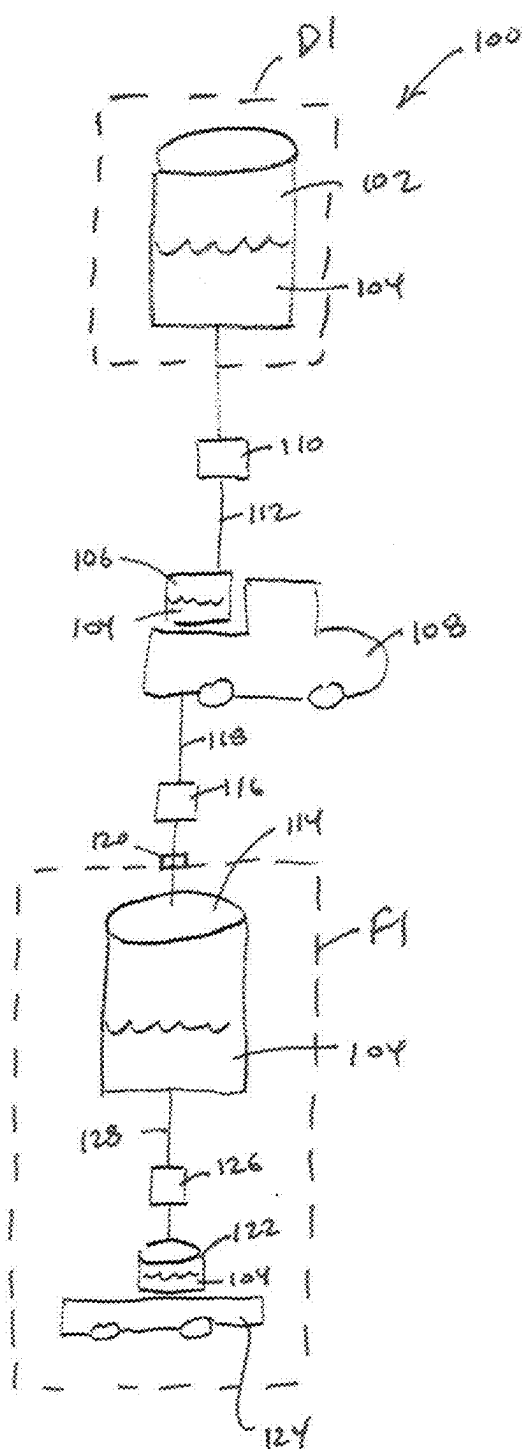


FIG. 1

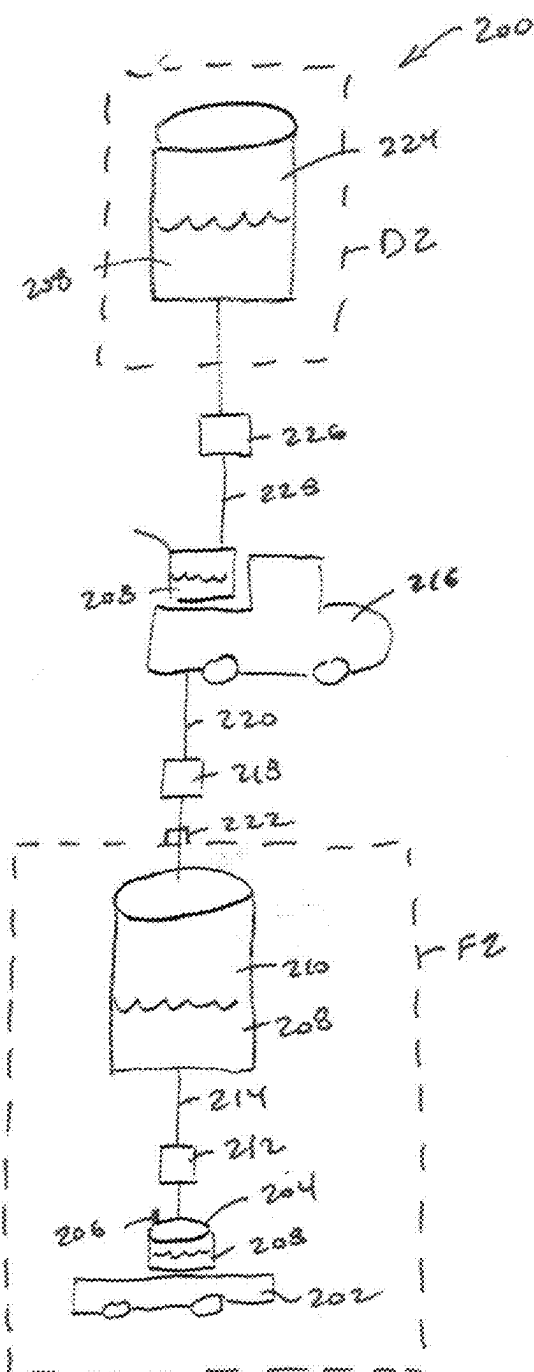


FIG. 2

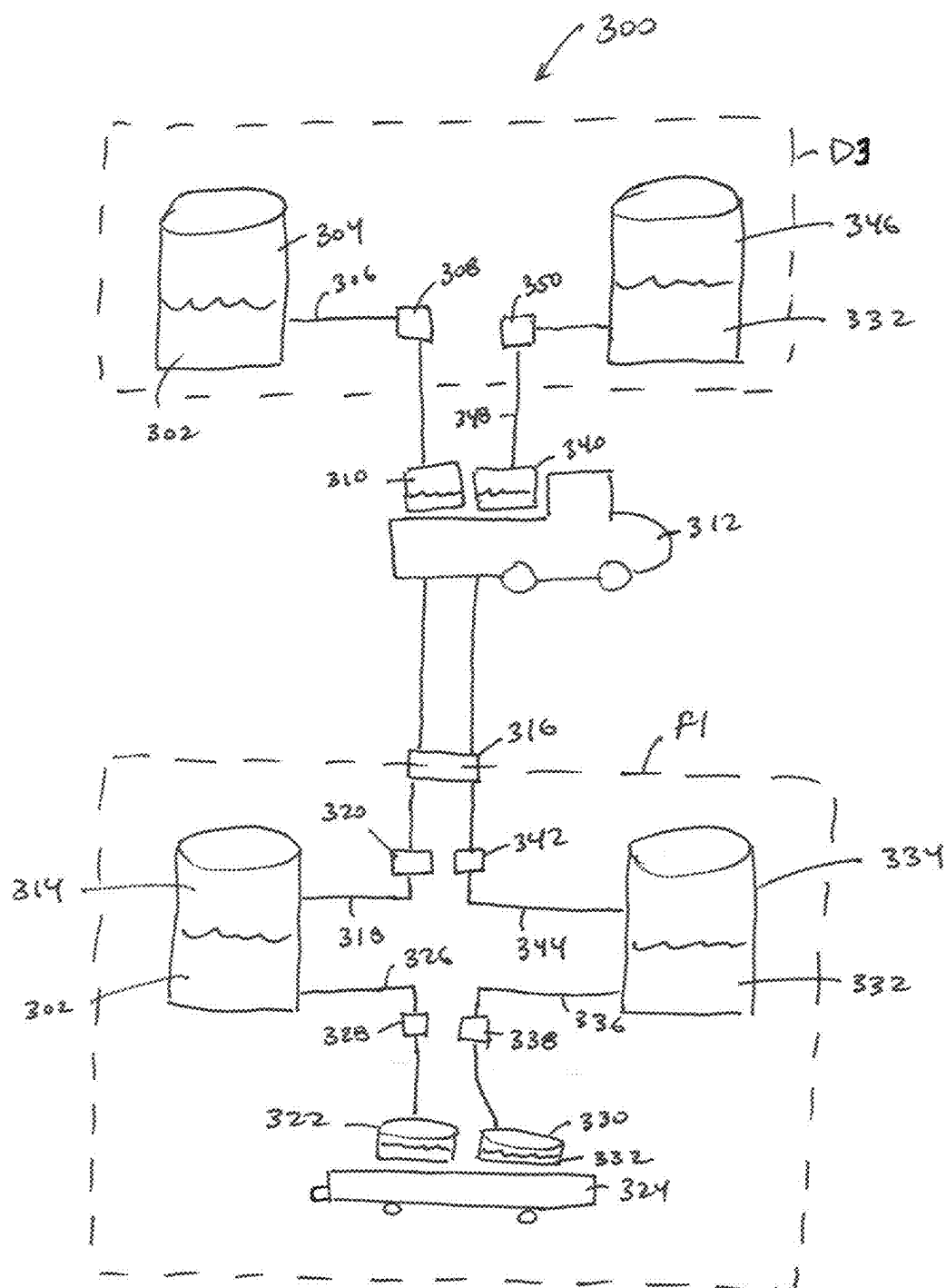


FIG. 3

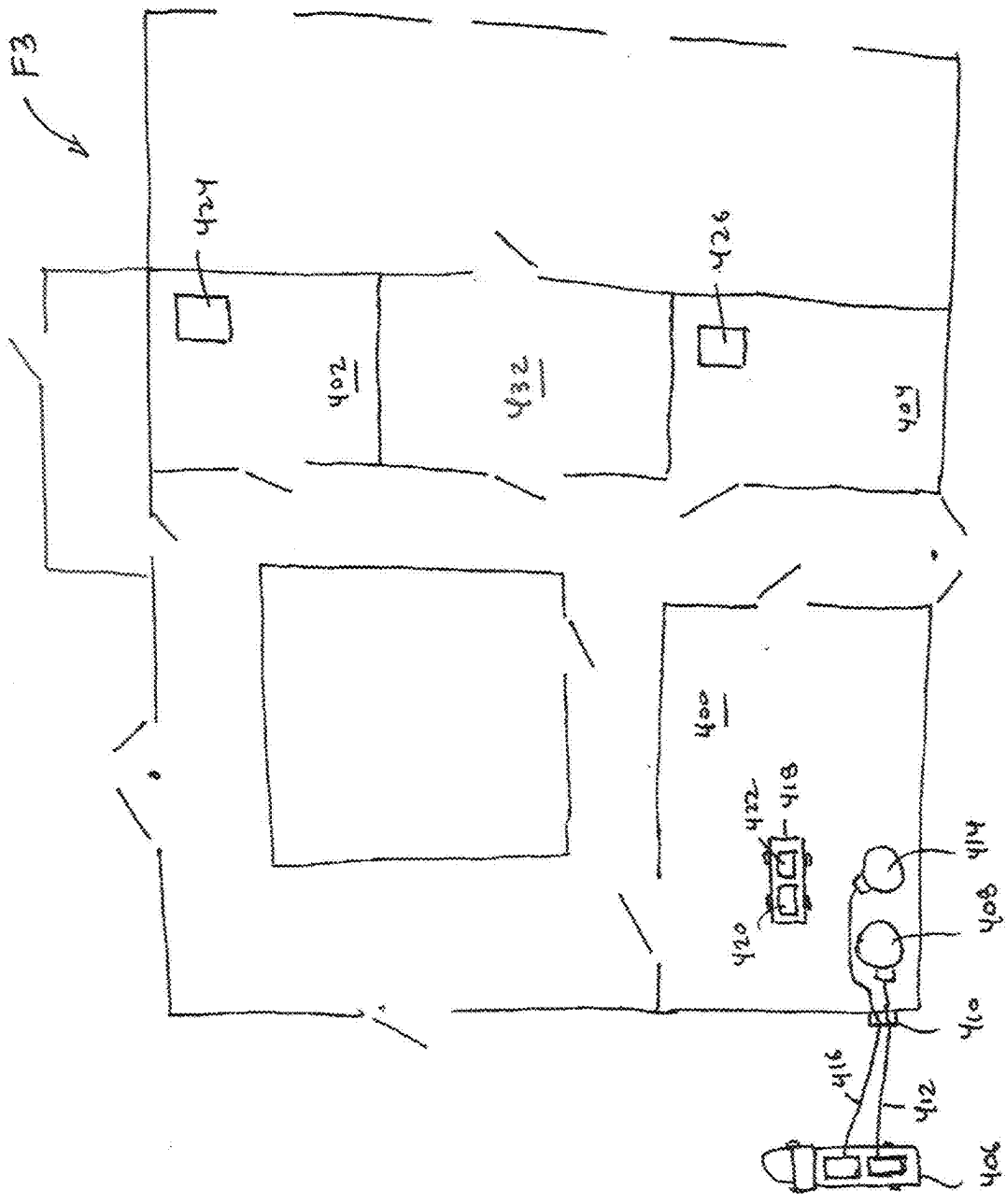


Fig. 4

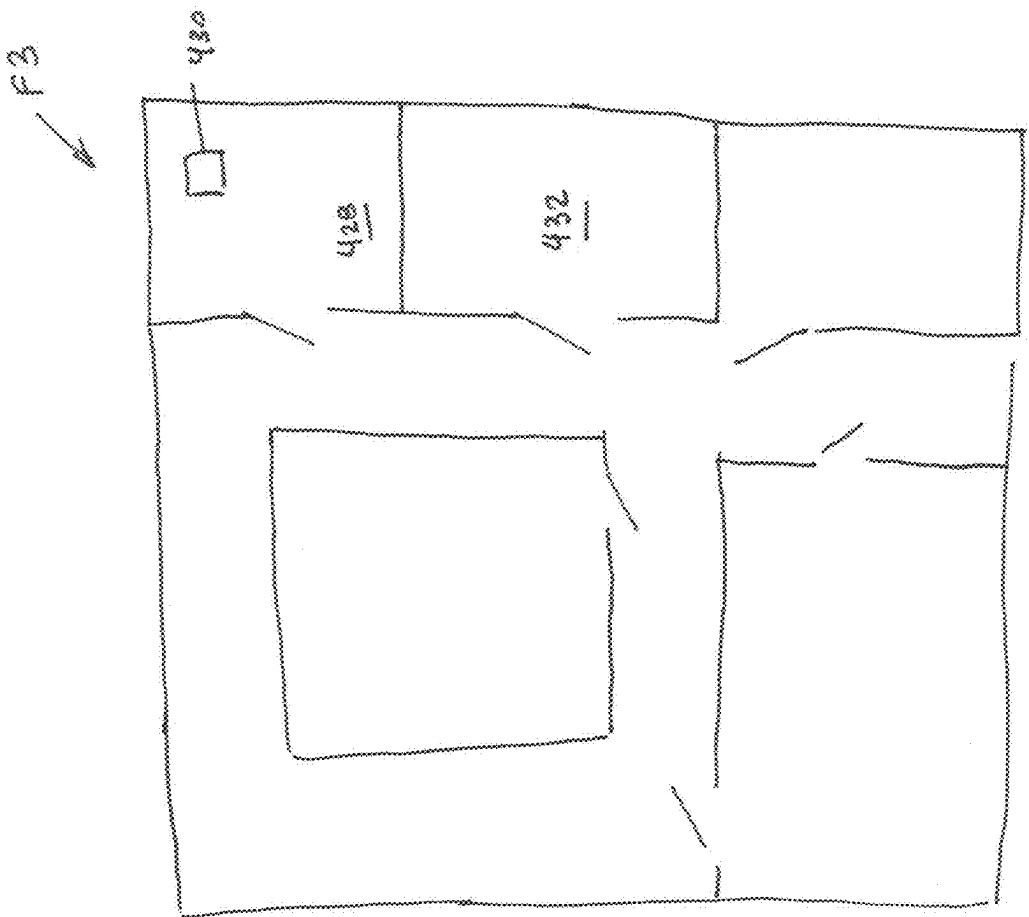


Fig. 5

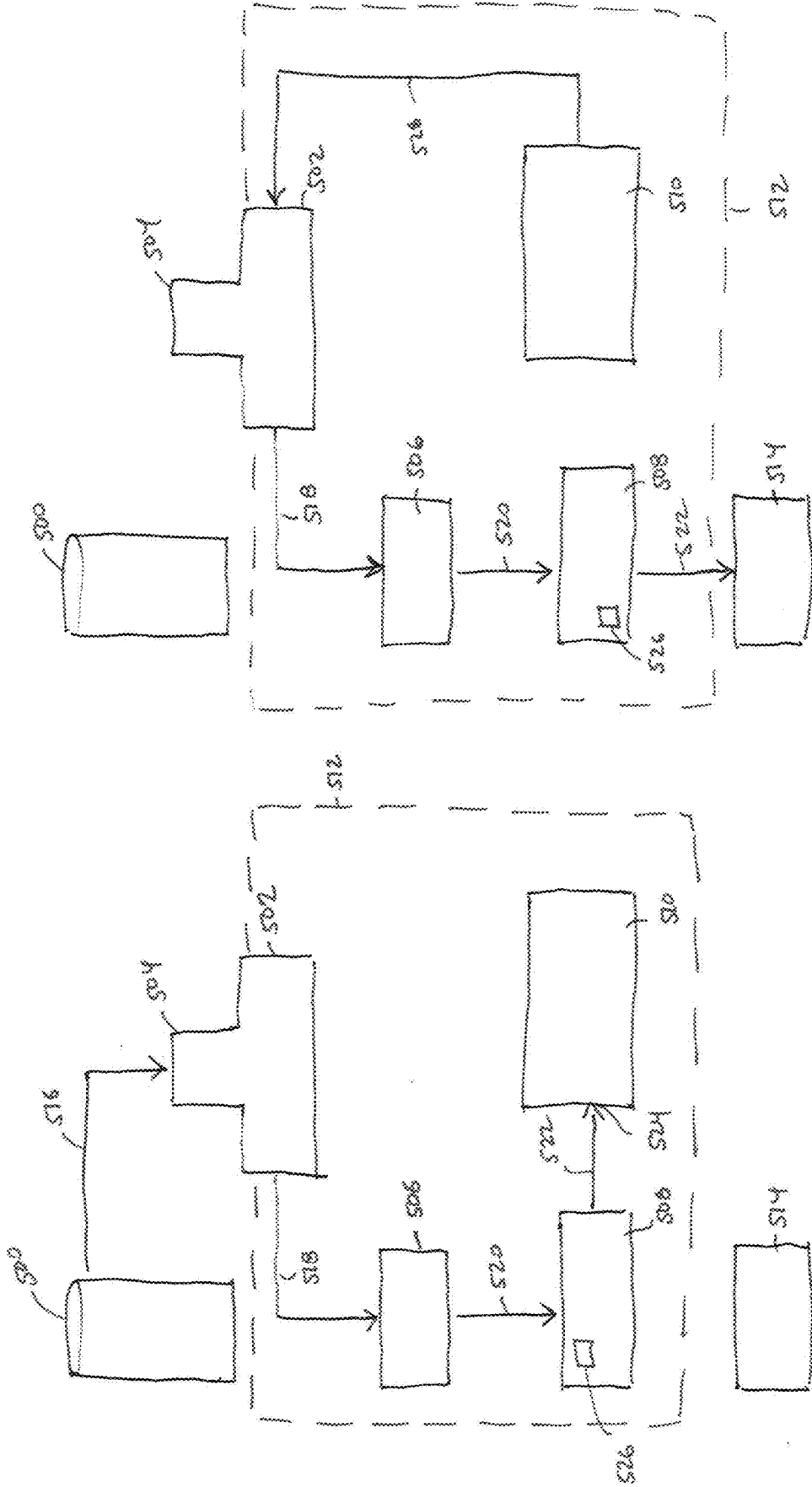


Fig. 6B

Fig. 6A

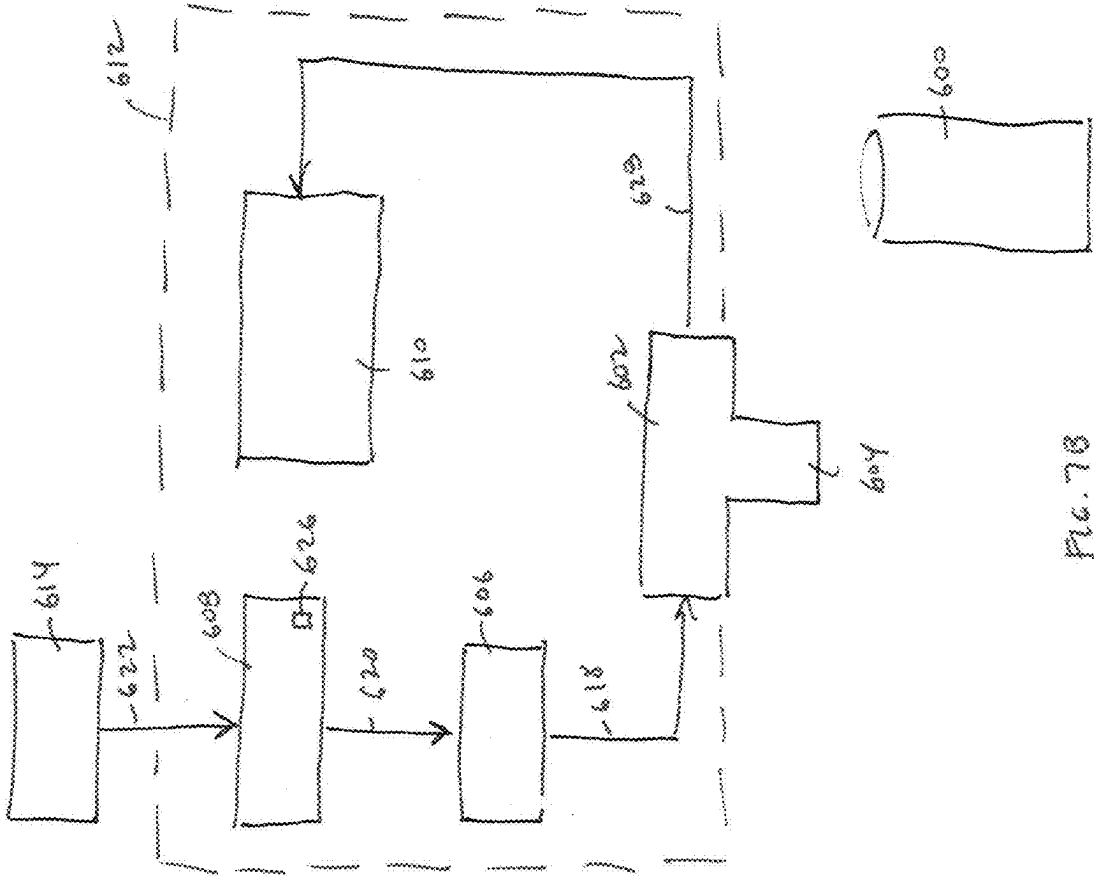


FIG. 7B

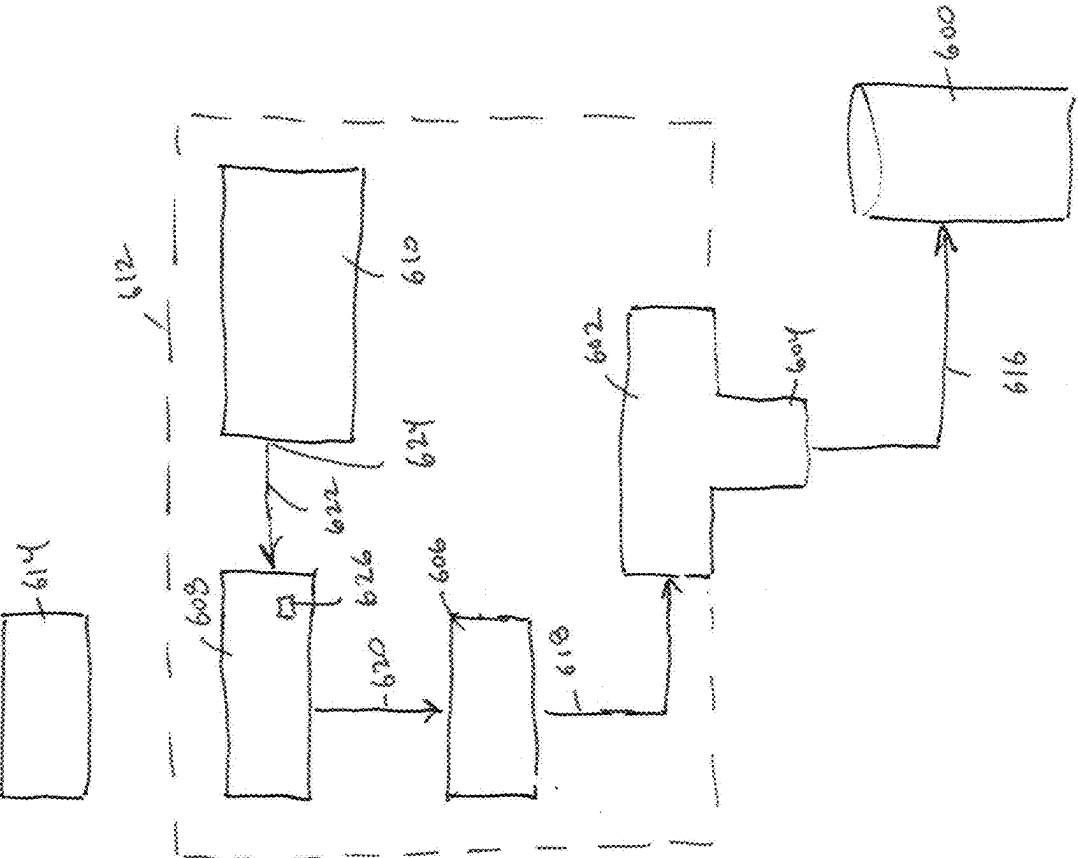


FIG. 7A

BULK COOKING OIL DISTRIBUTION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation application of U.S. patent application Ser. No. 18/376,743 filed Oct. 4, 2023, which is a continuation application of U.S. patent application Ser. No. 17/838,175 filed Jun. 11, 2022, which is a continuation application of U.S. patent application Ser. No. 17/103,594, filed Nov. 24, 2020, issued Jun. 21, 2022, as U.S. Pat. No. 11,364,836, which is a continuation application of U.S. patent application Ser. No. 16/505,443, filed Jul. 8, 2019, which is a continuation application of U.S. patent application Ser. No. 15/212,550, filed Jul. 18, 2016, issued on Jul. 9, 2019, as U.S. Pat. No. 10,343,584, and which is a continuation application of U.S. patent application Ser. No. 13/461,442, filed May 1, 2012, issued Jul. 19, 2016 as U.S. Pat. No. 9,392,907, the disclosures of each of which are hereby incorporated by reference herein.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to systems and methods for distributing cooking oil for use in the food services industry. As will be understood, distribution may also include collection of used cooking oil.

[0003] Throughout the world, a tremendous amount of cooking oil is used in the food services industry, primarily for deep frying foodstuffs. Traditionally, cooking oil was delivered to food preparation facilities in five-gallon containers. These heavy and somewhat awkward containers were stored in the facility and then carried by hand to deep fryers for filling with oil as necessary. This task was labor intensive and resulted in unnecessary waste packaging. It also resulted in a large amount of waste oil as residual oil often remained in the packaging after filling of the fryer despite the best efforts of the cooking staff.

[0004] Once the oil had gone through its useful life and was converted to used oil, the used oil was then spilled from the fryer into a five-gallon bucket for disposal in a remote dumpster, often located in an outdoor area of the cooking facility. Again, this was a labor intensive process that had many disadvantages. For one, the oil was typically removed from the fryer at the end of the business day and often was not permitted to sufficiently cool before an employee handled the oil, so transporting the hot oil by a tired employee often resulted in injury from burns. Additionally, general spills created an unsafe working area and slip and falls became problematic. Another detriment was that the remote dumpster became a haven for rodents and other unwanted pests. Finally, as used oil became more valuable, theft of the used oil from the storage location became a large issue.

[0005] Certain cooking oil distribution companies began using automated systems to alleviate the issues associated with the conventional method of supply and recovery. One such company is Oilmatic, LLC, which uses systems generally described in its U.S. Pat. No. 6,792,983, the disclosure of which is hereby incorporated by reference herein. In its popular commercial system, a fresh oil tank and used oil tank are provided in food preparation facility. Typically the tanks are provided indoors for security reasons. The tanks are piped independently through separate pumps to a hand

held distribution nozzle. One portion of the distribution nozzle is used to supply cooking oil to a fryer while a second portion is used to collect used cooking oil from the fryer. Operation is completely intuitive and no lifting of heavy oil is required.

[0006] In order to refill the fresh oil tank and remove used oil from the used oil tank, additional piping is provided. This piping typically terminates in a junction box located on an exterior wall of the cooking facility. A service truck may then attach piping to the junction box to fill the fresh tank and remove used oil from the used oil tank.

SUMMARY OF THE INVENTION

[0007] Although well received in the food services industry, one issue that has been discovered with respect to these systems is that some facilities, particularly those that are not strictly restaurants, have kitchens that are remote from external walls, have kitchens on a building level other than the first floor, or have several fryers that are remote from each other. While possible to provide piping to accommodate such a facility, it is often impractical from an aesthetics and/or cost standpoint.

[0008] It would therefore be beneficial to provide a bulk cooking oil distribution system that may adequately service such facilities.

[0009] The bulk cooking oil distribution systems and methods of the present invention are designed to overcome the deficiencies of conventional bulk cooking oil distribution systems, and are summarized generally below. Note that while various methods may be provided with steps in particular orders, those orders are not to be considered the only order in which the various methods may be conducted.

[0010] In one embodiment, a bulk cooking oil distribution method comprises receiving fresh cooking oil in a first fresh cooking oil tank; transferring at least a portion of the fresh cooking oil from the first fresh cooking oil tank to a second fresh cooking oil tank, the second fresh cooking oil tank being associated with a vehicle; transferring at least a portion of the fresh cooking oil from the second fresh cooking oil tank to a third fresh cooking oil tank located at a facility; transferring at least a portion of the fresh cooking oil from the third fresh cooking oil tank to a fourth fresh cooking oil tank, the fourth fresh cooking oil tank being associated with a first portable device; collecting used cooking oil in a first used cooking oil tank associated with a second portable device; transferring at least a portion of the used cooking oil from the first used cooking oil tank to a second used cooking oil tank located at the facility; transferring at least a portion of the used cooking oil from the second used cooking oil tank to a third used cooking oil tank, the third used cooking oil tank being associated with a vehicle; and transferring at least a portion of the used cooking oil from the third used cooking oil tank to a fourth used cooking oil tank.

[0011] The first fresh oil tank and the fourth used cooking oil tank may be located at a single depot.

[0012] The second fresh cooking oil tank and the third used cooking oil tank may be located on the same vehicle.

[0013] The first portable device and the second portable device may be the same portable device. If so, the portable device may comprise two pumps. Also where the portable device and the second portable device are the same portable devices, the portable device may comprise a propulsion mechanism. The propulsion mechanism may be powered by

one of electric, propane, or gasoline. Typically, the propulsion method will power wheels of the portable device.

[0014] The bulk cooking oil distribution method may further comprise transferring at least a portion of the fresh cooking oil from the fourth fresh cooking oil tank to a fryer in a kitchen of the facility. If so, the cooking oil distribution method may still further comprise transferring at least a portion of the fresh cooking oil from the fourth fresh cooking oil tank to a second fryer located in a second kitchen of the facility.

[0015] In the instance where the bulk cooking oil distribution method further comprises transferring at least a portion of the fresh cooking oil from the fourth fresh cooking oil tank to a fryer in a kitchen of the facility, the method may further comprise transferring at least a portion of the used cooking oil from the fryer in the facility to the first used cooking oil tank. This method may further comprise transferring at least a portion of the used cooking oil from a second fryer located in a second kitchen in the facility to the first used cooking oil tank.

[0016] In the various embodiments of the invention, the facility may be one of a stadium, arena, cruise ship, casino, shopping mall, convention hall, or multi-story building. While possible to provide piping to accommodate such a facility, it is often impractical from an aesthetics and/or cost standpoint.

[0017] In the bulk cooking oil distribution method, the facility may further comprise a kitchen at least partially enclosed by walls, wherein none of the walls is an exterior wall.

[0018] In the bulk cooking oil distribution method, the facility may further comprise a plurality of kitchens separated by walls, each kitchen having at least one fryer.

[0019] In the bulk cooking oil distribution method, the second fresh cooking oil tank may be integrated into the vehicle. In such case, the third used cooking oil tank may be integrated into the same vehicle.

[0020] In accordance with a further embodiment of the present invention, a cooking oil distribution method comprises the steps of receiving fresh cooking oil in a first fresh cooking oil tank; transferring at least a portion of the fresh cooking oil from the first fresh cooking oil tank to a second fresh cooking oil tank, the second fresh cooking oil tank being associated with a vehicle; transferring at least a portion of the fresh cooking oil from the second fresh cooking oil tank to a third fresh cooking oil tank located at a facility; providing instruction to transfer at least a portion of the fresh cooking oil from the third fresh cooking oil tank to a fourth fresh cooking oil tank, the fourth fresh cooking oil tank being associated with a first portable device; providing instruction to collect used cooking oil in a first used cooking oil tank associated with a second portable device; providing instruction to transfer at least a portion of the used cooking oil from the first used cooking oil tank to a second used cooking oil tank located at the facility; transferring at least a portion of the used cooking oil from the second used cooking oil tank to a third used cooking oil tank, the third used cooking oil tank being associated with a vehicle; and transferring at least a portion of the used cooking oil from the third used cooking oil tank to a fourth used cooking oil tank.

[0021] In the cooking oil distribution method, the first and second portable devices may be the same device.

[0022] In the cooking oil distribution method, the second fresh cooking oil tank and the third used cooking oil tanks may be integrated into the same vehicle.

[0023] In a still further embodiment of the present invention, a cooking oil distribution method comprises transferring at least a portion of fresh cooking oil from a fresh cooking oil tank to a second fresh cooking oil tank associated with a first portable device; transferring at least a portion of the fresh cooking oil from the second fresh cooking oil tank into a first fryer; collecting used cooking oil from the fryer in a first used cooking oil tank associated with a second portable device; and transferring at least a portion of the used cooking oil from the first used cooking oil tank to a second used cooking oil tank.

[0024] The cooking oil distribution method may further comprise cooking food products with the fresh cooking oil.

[0025] The cooking oil distribution method may further comprise transferring at least a portion of the fresh cooking oil from the second fresh cooking oil tank into a second fryer. If so, the first fryer and the second fryer may be located in separate kitchens. The separate kitchens may be on separate floors.

[0026] In the cooking oil distribution method, the first portable device and the second portable device may be the same device. The device may include two pumps. The two pumps may be connected to distribution lines that converge in parallel in a single hand held distribution nozzle.

[0027] In a further embodiment of the present invention, a bulk cooking oil distribution system comprises a first fresh cooking oil tank; a second fresh cooking oil tank and a first used cooking oil tank associated with a portable device; and a second used cooking oil tank.

[0028] The bulk cooking oil distribution system may further comprise a first pump and associated fresh oil piping for delivering fresh cooking oil from the first fresh cooking oil tank to the second fresh cooking oil tank and a second pump and associated used oil piping for delivering used cooking oil from the first used cooking oil tank to the second used cooking oil tank. The fresh oil piping and the used oil piping may be separate piping systems.

[0029] In the bulk cooking oil distribution system, the portable device may comprise a propulsion mechanism.

[0030] In the bulk cooking oil distribution system, the first fresh cooking oil tank and the second used cooking oil tank may be located adjacent to each other.

[0031] In the bulk cooking oil distribution system, the second fresh cooking oil tank is may be larger than the first used cooking oil tank.

[0032] In a still further embodiment of the present invention, there is provided a portable device for transferring cooking oil. The portable device comprises a first three-way valve, a first pump, a nozzle, and a fresh oil tank, with piping interconnecting the three-way valve to the pump, the pump to the nozzle, and the fresh oil tank to the three-way valve. An outlet tube of the nozzle may direct flow from the nozzle to the fresh oil tank or a fryer. The portable device also includes a second three-way valve, a second pump, and a used oil tank, with piping interconnecting the nozzle to the pump, the pump to the three-way valve, and the three-way valve to the used oil tank. An inlet tube of the nozzle may be submerged into the fryer or connected to the used oil tank. This used oil path, from the inlet tube to the used oil tank, is preferably completely separate from the fresh oil path.

Particularly, within the nozzle the two lines are generally parallel but do not interconnect.

[0033] In a cart service mode, the components may be utilized to direct flow of fresh oil from a bulk tank, to the first three-way valve via additional piping, then to the first pump, the nozzle, and finally to the fresh oil tank. The components may also be utilized in the cart service mode to direct flow of oil from the used oil tank to the nozzle, then to the pump and the second three-way valve, and finally through additional piping to a bulk used oil tank.

[0034] In a fryer service mode, the components may be utilized to direct fresh oil from the fresh oil tank to the first three-way valve, the first pump, the nozzle, and through the outlet tube of the nozzle to a fryer. The components may also be utilized to collect used oil from the fryer by transferring it from the fryer into the inlet tube of the nozzle, then through the nozzle itself to the pump, and then to the three-way valve and finally the used oil tank.

[0035] In an additional embodiment of the present invention, a cooking oil distribution device comprises a fresh oil tank, a three-way valve connected by piping to the fresh oil tank, a pump connected by piping to the three-way valve, and a nozzle connected by piping to the pump, the nozzle comprising an outlet tube connected to the piping between the pump and the nozzle, where the cooking oil distribution device is portable.

[0036] In such a device, cooking oil may be transferred from a bulk tank to the fresh oil tank by connecting piping from the bulk tank to the three-way valve, arranging the three-way valve to connect the piping from the bulk tank to the piping connected to the pump, arranging the outlet tube in fluid communication with the fresh oil tank, and energizing the pump, whereby cooking oil flows from the bulk tank, through the three-way valve, through the pump, through the nozzle, and through the outlet tube, into the fresh oil tank.

[0037] In such a device, cooking oil may be transferred from the fresh oil tank to a fryer by placing the outlet tube over the fryer, arranging the three-way valve to connect the piping from the fresh oil tank to the piping to the pump, and energizing the pump, whereby cooking oil flows from the fresh oil tank through the three-way valve, through the pump, through the nozzle, and through the outlet tube, into the fryer.

[0038] Such a device may further comprise a used oil tank, a second three-way valve connected by piping to the used oil tank, a second pump connected by piping to the three-way valve, and piping connecting the second pump to the nozzle, the nozzle further comprising an inlet tube connected to the piping between the pump and the nozzle.

[0039] In this modified device, used cooking oil may be transferred from a fryer by dipping the inlet tube into the used oil, arranging the second three-way valve to connect the piping from the second pump to the used oil tank, and energizing the second pump, whereby used cooking oil flows from the fryer through the inlet tube and the nozzle, through the pump and the three-way valve, and into the used oil tank.

[0040] In this modified device, used cooking oil may be transferred from the used oil tank to a bulk tank by placing the inlet tube in fluid communication with the used oil tank, arranging the three-way valve to connect the piping from the used oil tank to piping connected to the bulk tank, and energizing the second pump, whereby used cooking oil

flows from the used oil tank, through the inlet tube and the nozzle, through the pump and the three-way valve, and into the bulk tank.

[0041] This modified device may further comprise a platform and wheels. It may also further comprise a propulsion mechanism to propel at least one of the wheels.

[0042] In this modified device, the fresh oil tank and the used oil tank may be approximately 40 gallons each.

[0043] In this modified device, the fresh oil tank and the used oil tank may be sized to service approximately five to six fryers.

[0044] This modified device may be sized to be moved through a standard service hallway. The hallway may be in one of a stadium, arena, cruise ship, casino, shopping mall, convention hall, or multi-story building.

[0045] In the modified device, the piping may be configured such that fresh cooking oil does not mix with used cooking oil.

[0046] In a still further embodiment of the invention, a portable cooking oil distribution device comprises a fresh oil tank and a used oil tank with four modes of operation. A first mode of operation permits transfer of fresh cooking oil from a bulk tank to the fresh oil tank, a second mode of operation permits transfer of fresh cooking oil from the fresh oil tank to a fryer, a third mode of operation permits transfer of used cooking oil from a fryer to the used oil tank, and a fourth mode of operation permits transfer of used cooking oil from the used oil tank to a second bulk tank.

[0047] The portable cooking oil distribution device may further comprise a propulsion mechanism. The propulsion mechanism may be powered by one of electric, propane, or gasoline.

[0048] The portable cooking oil distribution device may further comprise a first pump associated with the first and second modes of operation and a second pump associated with the third and fourth modes of operation. In one embodiment, fresh oil in the first and second modes of operation does not mix with used oil in the third and fourth modes of operation.

[0049] In an additional embodiment of the invention, a portable cooking oil distribution device comprises a fresh oil tank, a used oil tank, and additional components adapted to collect used cooking oil from a fryer and to fill the fryer with fresh cooking oil.

[0050] Such a device may be sized to fit within a standard service hallway of a stadium, arena, cruise ship, casino, shopping mall, convention hall, or multi-story building.

BRIEF DESCRIPTION OF DRAWINGS

[0051] The subject matter regarded as the invention is particularly pointed out and distinctly claimed in the concluding portion of the specification. The invention, however, both as to organization and method of operation, together with features, objects, and advantages thereof, will be or become apparent to one with skill in the art upon reference to the following detailed description when read with the accompanying drawings. It is intended that any additional organizations, methods of operation, features, objects or advantages ascertained by one skilled in the art be included within this description, be within the scope of the present invention, and be protected by the accompanying claims.

[0052] With respect to the drawings, FIG. 1 is a schematic overview of a bulk cooking oil distribution system and associated methods in accordance with certain aspects of the invention;

[0053] FIG. 2 is a schematic overview of a further embodiment of the bulk cooking oil distribution system of the present invention;

[0054] FIG. 3 is a schematic overview of a still further embodiment of the bulk cooking oil distribution system of the present invention;

[0055] FIG. 4 is a first floor plan of a typical facility in which embodiments of the present invention may be utilized;

[0056] FIG. 5 is a second floor plan of the facility of FIG. 4;

[0057] FIG. 6A depicts an oil flow schematic of the fresh oil flow path in an exemplary cart while in the cart service mode;

[0058] FIG. 6B depicts an oil flow schematic of the fresh oil flow path in an exemplary cart while in the fryer service mode;

[0059] FIG. 7A depicts an oil flow schematic of the used oil flow path in an exemplary cart while in a cart service mode; and,

[0060] FIG. 7B depicts an oil flow schematic of the used oil flow path in an exemplary cart while in a fryer service mode.

DETAILED DESCRIPTION OF THE INVENTION

[0061] In the following are described the preferred embodiments of the bulk cooking oil distribution system in accordance with the present invention. In describing the embodiments illustrated in the drawings, specific terminology will be used for the sake of clarity. However, the invention is not intended to be limited to the specific terms so selected, and it is to be understood that each specific term includes all technical equivalents that operate in a similar manner to accomplish a similar purpose. Where like elements have been depicted in multiple embodiments, identical reference numerals have been used in the multiple embodiments for ease of understanding.

[0062] Throughout this disclosure reference is made to cooking oil stored in a tank and then transported to another tank. It is to be understood that when cooking oil is transferred, the same reference number may be utilized to represent a portion of that original cooking oil. For example, if the disclosure states that a first tank contains cooking oil and then states that cooking oil is transferred from the first tank to a second tank, it is to be understood that this includes less than the full amount of cooking oil, even if the same reference numeral is provided. Indeed, typically less than the entire amount of cooking oil may be transferred from tank to tank, particularly in relation to the fresh cooking oil supply. Another way to view this scenario is by considering the cooking oil to be a very small quantity, such that the entire “small quantity” of cooking oil from one tank is indeed transferred to the next tank.

[0063] It is also noted that various components may be described as connected, fluidly connected, or in fluid communication in relation to a flow path of oil. Where such terminology is used, it is to be understood in accordance with its broadest reasonable interpretation to include direct physical connections between the two components, or con-

nections through additional intervening components, so long as fluid may flow in the manner prescribed. Moreover, actual connection may not be necessary. For example, in relation to fresh oil flow from the outlet tube of the nozzle to a fryer, the fresh oil flow may merely be directed through the air above the fryer into the fryer while still being connected. Likewise, connection of the inlet tube of the nozzle may be achieved by dipping the inlet tube into used oil. Where it is discussed that piping is connected to the nozzle, it is to be understood that the piping may merely be contained within the nozzle and in actuality is fluidly connected with either the inlet tube or outlet tube.

[0064] With reference to FIG. 1, there is provided a schematic view of a bulk cooking oil distribution system 100 and associated methods in accordance with certain aspects of the invention. In accordance with this first embodiment, only the supply side 100 is shown. In this supply side embodiment 100, a first fresh cooking oil tank 102 is provided at an oil distribution depot D1. Typically, this first fresh cooking oil tank 102 will be of a capacity of between 1,000 and 5,000 gallons, but may be more or less depending on the needs of the distributor. Such tanks are commonly provided in metals such as stainless steel or aluminum, or plastics such as polypropylene, and should be FDA and HACCP approved. Of course, other materials may be used.

[0065] Within the first fresh cooking oil tank 102 is a supply of fresh cooking oil 104. This supply is commonly provided by a bulk cooking oil manufacturer and is brought to the distribution depot D1 via a supply vehicle (not shown). From the supply vehicle the fresh cooking oil 104 is conveyed into tank 102 via a transfer pump and associated piping provided either in conjunction with the supply vehicle or the depot D1.

[0066] When it becomes desirous to transport fresh cooking oil 104 from the depot D1 to a facility that uses cooking oil F1, fresh cooking oil may be transferred to a second fresh cooking oil tank 106 which is associated with a vehicle 108. Depending on the oil shipment size required, this association may be a separate tank located in the rear of a pickup truck or box truck, or may be a completely integrated tank in a tanker truck. Typically, this second fresh cooking oil tank will have a capacity of between 100 and 3,000 gallons and will be either metal or plastic if a separate tank in a pickup truck or box truck, or metal if integrated into a tanker truck. The tank 106 should also be FDA and HACCP approved.

[0067] In order to convey the fresh cooking oil 104 from the first fresh cooking oil tank 102 to the second fresh cooking oil tank 106, a transfer pump 110 and piping 112 are employed. There are various methods for handling this operation that are well known to the skilled artisan. For present purposes, it is to be understood that the transfer pump 110 may be integrated into the vehicle or may be a fixture in the depot D1. If a fixture, transfer pump 110 and piping 112 may be the same transfer pump and piping used to transfer fresh oil from the supply vehicle to the first fresh cooking oil tank 102.

[0068] Vehicle 108 is then utilized to transport the fresh cooking oil 104 to a third fresh cooking oil tank 114 located at a facility that uses cooking oil F1. Although any facility that uses cooking oil may utilize the invention herein, the facilities where this type of system are primarily useful typically include, but are not limited to, restaurants, grocery stores, stadiums and arenas, shopping malls, convention centers, commercial buildings, sailing vessels such as cruise

ships, and the like. In general, the most appropriate facilities for this technology are those where conventional forms of cooking oil distribution are impractical because of the distance between any kitchen and an exterior wall (whether a horizontal distance or vertical distance) or because the facility utilizes a number of kitchens where each requires cooking oil and where each may be remote from each other.

[0069] Upon arrival of the vehicle **108** at the facility **F1**, fresh cooking oil **104** is conveyed to third fresh cooking oil tank **114**. This conveyance is typically achieved by utilizing a transfer pump **116** and associated piping **118**. Commonly, this transfer pump **116** and piping **118** forms a portion of vehicle, and therefore travels with vehicle. However, that need not be true and the transfer pump **116** and/or piping **118** may remain at the facility **F1**. As is commonly known in the industry for delivery of fresh cooking oil, an outer wall of facility **F1** may include a junction box **120** for placing second fresh cooking oil tank **106** in fluid communication with third fresh cooking oil tank **114**. In such case, the piping upstream of the junction box **120** will be associated with the vehicle **108** and the piping downstream of the junction box will be associated with the facility **F1**.

[0070] It will be appreciated that third fresh oil tank **114** is preferably located at the facility **F1** in reasonable proximity to an exterior wall. In this manner, the run of piping need not excessively interfere with the aesthetics of the facility **F1** nor be too costly to install. Moreover, by maintaining a reasonable run of piping, transfer pump **116** may be kept to a minimal power rating, also serving to keep costs of the system lower.

[0071] The capacity of fresh cooking oil tank **114** will depend generally on the needs of the facility **F1** for which it is used. For example, a cruise ship that may not be able to refill for weeks at sea may require a large tank on the order of 3,000 gallons while a shopping mall, that many have various kitchens in a food court but which can be refilled more often, may only require 1,000 gallons. Facilities such as restaurants that have even fewer number of kitchens, or one remote kitchen, may only require 100 gallons. Typically, however, this second fresh cooking oil tank **114** will have a capacity of between 100 and 3,000 gallons and will be either metal or plastic and should be FDA and HACCP approved.

[0072] Once the fresh oil **104** is transferred to the third fresh cooking oil tank **114** in facility **F1**, it is available for use by facility **F1** personnel as instructed. Primarily, this manner will be for use in fryers in one or more kitchens. To transfer the fresh cooking oil **104** to the fryers, the fresh cooking oil is transferred to a fourth fresh cooking oil tank **122**, where the fourth fresh cooking oil tank is associated with a first portable device **124**.

[0073] Preferably, the fourth fresh cooking oil tank **122** of the first portable device **124** has a capacity of approximately 40 gallons. This size strikes a balance between portability and useful capacity as 40 gallons of fresh oil can be pushed by hand but is still enough to fill 5-6 standard sized fryers. Obviously, the fourth fresh cooking oil tank **122** may have other capacities, for example 10 gallons to 50 gallons, or more.

[0074] The fourth fresh cooking oil tank **122** may be configured from plastic or metal and should be FDA and HACCP approved.

[0075] Additional details of exemplary portable devices **124** will be provided below, however it should be noted that the portable device preferably includes a transfer pump **126**

and associated piping **128**. It is through this transfer pump **126** and associated piping **128** that fresh cooking oil **104** is conveyed from the third fresh cooking oil tank **114** to the fourth fresh cooking oil tank **122**. In other embodiments, a pump **126** that is not associated with the portable device **124** may be utilized. In other embodiments, the fourth fresh oil tank **114** may have the ability to open at the top (or may be open at the top) while the third fresh oil tank **114** is provided with a drain valve (not shown), whereby the fourth fresh cooking oil tank **122** may be positioned below the drain valve with the top of the fourth fresh cooking oil tank opened to drain fresh cooking oil directly into the fourth fresh cooking oil tank.

[0076] Once the fourth fresh cooking oil tank **122** is filled, or partially filled, with fresh cooking oil **104**, portable device **124** can be moved from the location of the fourth fresh cooking oil tank to one or more remote kitchens. At each of the kitchens, fresh cooking oil **104** is transferred from the fourth fresh cooking oil tank **122** to one or more fryers located in the respective kitchen. As will be discussed below, this transfer is preferably conducted via a transfer pump and piping, but may utilize other means as well.

[0077] It is noted that the portable device **124** may be a simple platform **125** with wheels **127** that is pushed manually. More intricate devices may power the cart with a propulsion mechanism **129**, such as those powered by electric, propane, or gasoline. In this case, the portable device **124** may be akin to a golf cart-type device or a powered platform truck. In any event, the portable device **124** is preferably small enough that it can fit within the confines of a facility, including accessing remote kitchen areas.

[0078] Once the fresh cooking oil **104** is transferred to one or more fryers it may be used in the ordinary course by the kitchen staff whereby it is converted into used cooking oil.

[0079] In accordance with certain embodiments of the present invention, including those provided in FIG. 2, a bulk cooking oil distribution system **200** comprises apparatuses for the collection of used cooking oil. It should be noted that in a preferred embodiment, the apparatuses shown in FIG. 1 and FIG. 2 are included in the same bulk cooking oil distribution system, as shown in FIG. 3.

[0080] As shown in FIG. 2, a second portable device **202** may be provided with a first used cooking oil tank **204**. Like the first portable device **124**, the second portable device **202** may be a simple platform with wheels that is pushed manually. More intricate devices may power the cart with a propulsion mechanism, such as those powered by electric, propane, or gasoline. In this case, the portable device **202** may be akin to a golf cart-type device or a powered platform truck. In any event, the portable device **202** is preferably small enough that it can fit within the confines of the facility, including accessing remote kitchen areas. In some embodiments of the invention, the first and second portable devices **124**, **202** may be the same portable device configured with two separate tanks, a fresh cooking oil tank and a used cooking oil tank. Such a portable device is shown in FIG. 3.

[0081] Preferably, the first used cooking oil tank **204** of the second portable device **202** has a capacity of approximately 40 gallons. This size strikes a balance between portability and useful capacity as 40 gallons of used oil can be pushed by hand but is still enough to collect used cooking oil from 5-6 standard sized fryers. Of course, the first used cooking oil tank **204** may have other capacities, for example 10 gallons to 50 gallons, or more.

[0082] In the case where the fourth fresh cooking oil tank 122 and the first used cooking oil tank 204 are associated with the same portable device, it is preferred that the fourth fresh cooking oil tank and the first used cooking oil tank have approximately the same volume, or at least be sized for servicing the same number of fryers. If there is any discrepancy in the volume, it is preferred that the fourth fresh cooking oil tank 122 be of a slightly larger capacity, approximately 5-20%, to account for cooking oil loss during the cooking process. In this regard, it is well known that some of the cooking oil used in cooking remains with the cooked food such that used cooking oil volume is less than the fresh cooking oil volume placed in any fryer.

[0083] The first used cooking oil tank 204 may be provided in metals such as stainless steel or aluminum, or plastics such as polypropylene, and should be FDA and HACCP approved. Of course, other materials may be used. To account for potential filling with used cooking oil at an elevated temperature, it is preferred that the first used cooking oil tank 204 be made of metal and possibly be insulated to protect other components of the second portable device 202 and the second portable device itself. It is also preferred that the first used cooking oil tank 204 be provided with a vent 206 to permit air exchange with the atmosphere to prevent excessive pressure build up.

[0084] Used cooking oil 208 from the various fryers in the facility F2 may be collected in the first used cooking oil tank 204 by being spilled directly into the first used cooking oil tank via a drain spout in the fryer or collected using a hand held distribution nozzle that may be associated with the second portable device.

[0085] Once the used cooking oil 208 from the facility F2 is collected in the first used cooking oil tank 204, the first portable device 202 may be moved to a second used cooking oil tank 210. If the facility F2 includes a fresh oil supply system 100, such that F1 is F2, then the second used cooking oil tank 210 is preferably located at the facility F2 adjacent the third fresh cooking oil tank 114. The used cooking oil 208 may then be transferred from the first used cooking oil tank 204 to the second used cooking oil tank 210 via a transfer pump 212 and associated piping 214. The transfer pump 212 and associated piping 214 may be integrated into the portable device 202 or may be separate.

[0086] The second used cooking oil tank 210 is typically of a capacity of between 1,000 and 3,000 gallons, but may be more or less depending on the needs of the facility. Commonly, the second used oil tank 210 will be sized approximately equal to, or slightly less than, the size of the third fresh cooking oil tank 114, if so provided. Such tanks are commonly provided in metals such as stainless steel or aluminum, or plastics such as polypropylene, and should be FDA and HACCP approved. Of course, other materials may be used. Because the used oil 208 will typically cool before being transferred to the second used oil tank 210, one typically does not have to account for elevated oil temperatures in the tank.

[0087] The used cooking oil collected in the second used cooking oil tank 210 may be transferred to a third used cooking oil tank 215 associated with a vehicle 216. The vehicle 216 may have the third used cooking oil tank 215 integrated therein, for example in a tanker truck, or may have a separate tank mounted to the vehicle, for example the rear of a pickup truck or box truck. Typically, this third used cooking oil tank will have a capacity of between 100 and

3,000 gallons and will be either metal or plastic if a separate tank in a pickup truck or box truck, or metal if integrated into a tanker truck.

[0088] To convey used cooking oil from the second used cooking oil tank 210 to the third used cooking oil tank 215, a pump 218 and associated piping 220 is utilized. The pump 218 and piping 220 may be associated with the vehicle 216 or may remain in the facility F2. Preferably, piping that terminates in a junction box 222 located on an exterior wall of the facility F2 is provided. A vehicle may then attach piping to the junction box 222 to draw the used cooking oil 208 from the second used cooking oil tank 210, either with a pump associated with the vehicle or a pump associated with the facility.

[0089] Used cook oil 208 contained in the third used cooking oil tank 214 may be brought by the vehicle 216 directly to a used cooking oil renderer, or preferably to a depot D2. In a most preferred system, depot D1 and depot D2 are the same depot. Located at depot D2 is a fourth used cooking oil tank 224. This fourth used cooking oil tank 224 is typically of a capacity of between 1,000 and 5,000 gallons, but may be more or less depending on the needs of the distributor. Such tanks are commonly provided in metals such as stainless steel or aluminum, or plastics such as polypropylene, and should be FDA and HACCP approved. Of course, other materials may be used. Once collected the fourth used cooking oil tank 224, the used cooking oil 208 may be rendered on site for other use or may be transferred to an off-site renderer.

[0090] In order to convey the used cooking oil 208 to the fourth used cooking oil tank 224 from the third used cooking oil tank 214, a pump 226 and piping 228 are used. The pump 226 and piping 228 may form part of the vehicle 216 or may be part of the depot D2. Where depot D1 and depot D2 are the same depot, pump 226 and pump 110 may be the same pump.

[0091] Typically most of the activities of the bulk cooking oil system will be conducted by a single entity, or at the instruction of a single entity. For example, the oil distribution entity will either own or lease the depot D1. Its personnel will be responsible for receiving fresh cooking oil in the first fresh cooking oil tank and thereafter be responsible for conveyance of the fresh cooking oil all the way to the third fresh cooking oil tank at facility F1. Conveyance of the fresh cooking oil into the fourth fresh cooking oil tank and any fryer will then be at the instruction of the entity and typically conducted by facility F1 personnel. Likewise, collection of used cooking oil and conveyance to the second used cooking oil tank will typically be at the instruction of the entity and conducted by facility F2 personnel. Finally, entity personnel will typically be directly responsible for conveyance of the used cooking oil from the second used cooking oil tank to the fourth used cooking oil tank. It is noted that use of the term instruction is not to be limited to mere instruction but may also include direction and control.

[0092] FIG. 3 depicts a typical bulk cooking oil distribution system 300, which includes both supply and recovery of cooking oil. It will be appreciated that the majority of the operations of the bulk cooking oil distribution system 300 operate in much the same manner as the two systems 100, 200 previously discussed. However, the system 300 will be briefly discussed below.

[0093] Once again, fresh cooking oil 302 is delivered to a first fresh cooking oil tank 304 at a depot D3. From there it

is conveyed through piping **306** and a pump **308** to a second fresh cooking oil tank **310** forming part of a vehicle **312**. The vehicle **312** then brings the fresh cooking oil **302** to the facility **F1** that is desirous of the fresh oil, and conveys that oil to a third fresh cooking oil tank **314**. The conveyance is through a junction box **316** on the outer wall of the facility **F1** and incorporates piping **318** and a pump **320**. From there, the fresh oil **302** is moved to a fourth fresh cooking oil tank **322** mounted on a cart **324**. That conveyance is through piping **326** and a pump **328**.

[0094] The cart **324** may then be brought to the locations of fryers within the facility **F1** to transfer the fresh cooking oil to the fryers. In this embodiment, the cart **324** also includes a used cooking oil tank **330** that can collect used cooking oil **332**. That used cooking oil **332** can then be brought to a second used cooking oil tank **334** located in the facility, preferably near an external wall, and preferably adjacent to the third fresh cooking oil tank **314**. Conveyance of used cooking oil **332** to the second used cooking oil tank is achieved via piping **336** and a pump **338**.

[0095] Vehicle **312** may then be utilized to collect the used cooking oil **332** in a third used cooking oil tank **340** associated with the vehicle. The used cooking oil **332** is conveyed to the vehicle **312** through the junction box **316** and a pump **342** with associated piping **344**. Preferably, the vehicle **312** will pick up used cooking oil **332** during the same trip that it delivers fresh cooking oil **302**.

[0096] Once loaded with used cooking oil **332**, the vehicle will return to the depot **D3** for delivery of the used cooking oil to a fourth used cooking oil tank **346**. This delivery is made through piping **348** and a pump **350**.

[0097] FIG. 4 depicts a first floor plan of an exemplary facility **F3** that may use cooking oil. The facility **F3** includes a number of rooms and connecting service hallways, including a storage room **400** and kitchens **402**, **404**. It will be appreciated that although the kitchens **402**, **404** are located on external walls, they are remote from each other and therefore piping cooking oil using conventional methods is not practical.

[0098] Instead, the facility **F3** is supplied with cooking oil using systems and methods of the present invention. Here, a vehicle **406** delivers fresh cooking oil (not shown) to the facility **F3** and conveys that oil to a first fresh cooking oil tank **408** through a junction box **410** and piping **412**. It is noted here that the vehicle **406** may also collect used cooking oil from a second used cooking oil tank **414** through the junction box **410** and piping **416**.

[0099] Once the fresh cooking oil is supplied to the first fresh cooking oil tank **408**, the oil is available for use by the facility. To convey the oil to the remote kitchens **402**, **404**, a portable cart **418** is utilized. The portable cart includes a second fresh cooking oil tank **420** for this purpose as well as a first used cooking oil tank **422** for collection of used cooking oil.

[0100] It will be appreciated that oil is conveyed from the first fresh cooking oil tank **408** to the second fresh cooking oil tank **420** of the cart by various means previously discussed. The cart **418** may then be pushed or driven to the remote kitchens **402**, **404** to fill fryers **424**, **426** located in the kitchens. During the same trip that the fryers **424**, **426** are filled, used cooking oil from the fryers can be deposited into the first used cooking oil tank **422** of the cart. The cart can

then be returned back to the supply room **400** where the used oil is conveyed to the second used cooking oil tank **414** for later pickup.

[0101] It will also be appreciated that there may be more than two kitchens provided in any facility, and those kitchens may also be on separate floors. For example, FIG. 5 depicts a second floor plan of the facility **F3** of FIG. 4. In FIG. 5, there is shown a third kitchen **428** with a third fryer **430** located on this second floor. In order to service this third kitchen **428** and fryer **430** with oil from storage room **400**, there is provided an elevator **432**. The cart **418** may therefore travel in the elevator between the first and second floors of facility **F3** to service all of the kitchens.

[0102] Preferably, the cart **418** is sized to easily maneuver through the facility **F3**, including standard sized service hallways and doorways, and is relatively compact. This compact configuration is partially derived through the use of minimal components in a novel manner. Some of the components can be those shown and described in U.S. Pat. No. 6,792,983, including the nozzle.

[0103] The cart generally includes two modes of operation. In a cart service mode, fresh cooking oil may be delivered from a bulk tank to a fresh oil tank associated with the cart. In the meantime, used cooking oil may be removed from the used oil tank on the cart and transferred to a bulk used cooking oil tank. Both of these activities will typically be conducted in the direct vicinity of the bulk tanks.

[0104] In a fryer service mode, typically entered into at the kitchen sites, the cart is used to service fryers. Here, fresh cooking oil from the fresh cooking oil tank associated with the cart may be pumped to a fryer. Additionally, used cooking oil may be removed from the fryer and pumped into the used cooking oil tank associated with the cart. In both cases, the nozzle of U.S. Pat. No. 6,792,983 may be utilized.

[0105] FIG. 6A depicts an oil flow schematic of the fresh oil flow path in an exemplary cart while in the cart service mode. Shown in FIG. 6A is a bulk tank **500** for the storage of fresh cooking oil. Also shown are a three-way valve **502** with quick disconnect **504**, pump **506**, nozzle **508**, and fresh oil tank **510**, all of which are associated with the cart **512**. Also shown is a fryer **514** located in a kitchen.

[0106] In order to transfer fresh cooking oil from the bulk tank **500** to the fresh oil tank **510**, piping **516**, preferably in the form of a flexible hose, is connected from the bulk tank to the quick disconnect **504** of the cart **512**. For purposes of this connection it is preferred that the piping **516** and the quick disconnect **504** have characteristics that are specific to that connection, such as size and configuration, so that the piping is not accidentally connected to the quick disconnect associated with the used oil, as will be discussed below. In this mode of operation the three-way valve **502** is set up, either manually or electronically, to connect piping **516** to piping **518** configured between the three-way valve and the pump **506**. The pump **506** is in turn connected by piping **520** to the nozzle **508**. Piping **520** in turn is connected to the outlet tube **522** of the nozzle **508**. Preferably this piping is hard piped and somewhat permanently connected. In this mode of operation, the outlet tube **522** is placed within a port **524**, or opening, within the cart **512** that leads to the fresh oil tank **510**.

[0107] Therefore, when it is desired to fill the fresh oil tank **510** with fresh cooking oil, an operator may plug piping **516** into the quick disconnect **504** and the outlet tube **522** of the nozzle **508** into the port **524**. The operator then ensures

that the three-way valve **502** is properly oriented and energizes the pump **506** using controls **526** on the nozzle **508**. It will be appreciated that the pumps described herein may be powered by battery or A/C current.

[0108] It will be appreciated that various means for ensuring that the fresh oil tank **510** does not overflow may be incorporated. Such means may include a visual level meter, electronic shutoff, or flow timer. Additional means within the purview of one skilled in the art may also be utilized.

[0109] As previously discussed, piping **516** is preferably in the form of flexible hose. While piping **518** and **520** may also be formed from flexible hose, it is preferred that such piping be configured for relatively permanent connection and therefore may be hard piped, such as with metals. These metals may be stainless steel.

[0110] FIG. 6B depicts an oil flow schematic of the fresh oil flow path in an exemplary cart while in the fryer service mode. Shown in FIG. 6B is the bulk tank **500** for the storage of fresh cooking oil. Also shown are the three-way valve **502** with quick disconnect **504**, pump **506**, nozzle **508**, and fresh oil tank **510**, all of which are associated with the cart. Likewise, the fryer **514** is shown in a kitchen.

[0111] In order to transfer fresh cooking oil from the fresh oil tank **510** of the cart **512**, additional piping is provided. In this case, piping **518** and **520** remain in their respective positions between the three-way valve **502** and pump **506**, and between the pump and nozzle **508**, respectively. The outlet tube **522** of the nozzle **508** is not positioned within the port **524** of the cart **512** but rather is positioned above the fryer **514** to dispense oil therein. Furthermore, piping **528** is fixed between the fresh oil tank **510** and the three-way valve **502**, as shown. Lastly, piping **516** (FIG. 6A) between the bulk tank **500** and quick disconnect **504** is no longer necessary as presumably the cart **512** would be moved to a kitchen while in the fryer service mode.

[0112] In this mode of operation the three-way valve **502** is set up, either manually or electronically, to connect piping **528** from the fresh oil tank **510** to piping **518** configured between the three-way valve and the pump **506**. Therefore, the connection to the quick disconnect **504** is closed off. The pump **506** is in turn connected by piping **520** to the nozzle **508**. Piping **520** is connected to the outlet tube **522** of the nozzle **508**.

[0113] When it is desired to fill the fryer **514** with fresh cooking oil, an operator may place the outlet tube **522** of the nozzle **508** above the fryer. The operator then ensures that the three-way valve **502** is properly oriented and energizes the pump **506** using controls **526** on the nozzle **508**. The pump **506** will then draw fresh oil from the fresh oil tank **510** through piping **528**, three-way valve **502**, and piping **518**. The pump **506** will then push that fresh oil through piping **520**, nozzle **508**, and outlet tube **522** into the fryer **514**.

[0114] FIG. 7A depicts an oil flow schematic of the used oil flow path in an exemplary cart while in a cart service mode. Note that the cart shown in FIG. 7A may be the same cart shown in FIG. 6A, or may be a separate cart. Preferably, it is the same cart.

[0115] Shown in FIG. 7A is a bulk tank **600** for the storage of used cooking oil. Also shown are a three-way valve **602** with quick disconnect **604**, pump **606**, nozzle **608**, and used oil tank **610**, all of which are associated with the cart **612**. Also shown is a fryer **614** located in a kitchen.

[0116] In order to transfer used cooking oil from the used oil tank **610** to the bulk tank **600** piping is provided. In the

cart itself, this piping comprises piping **620** connecting the nozzle **608** to the pump **606** and piping **618** connecting the pump to the three-way valve **602**. Preferably this piping is hard piped and somewhat permanently connected. In the meantime, inlet tube **622** of the nozzle **608** may be placed in a port **624** to connect the used oil tank **610** to the nozzle **608**.

[0117] From the quick disconnect **604** of the three-way valve **602**, additional piping **616** connects to the bulk tank **600**. This piping may be in the form of flexible hose. For purposes of the connection between the quick disconnect **604** and the bulk tank **600**, it is preferred that the piping **616** and the quick disconnect **604** have characteristics that are specific to that connection, such as size and configuration, so that the piping is not accidentally connected to the quick disconnect associated with the fresh oil. In one exemplary embodiment, the quick disconnect **504** on the fresh oil side is a male connection while the quick disconnect **604** on the used oil side is a female connection. Others configurations such as different sized connection diameters may also be utilized.

[0118] When it is desired to transfer used oil from the used oil tank **610** to the bulk tank **600**, the operator will move the cart **612** to the location of the bulk tank **600**. The operator will then insert the inlet tube **622** of the nozzle **608** into the port **624**. This effectively connects the nozzle **608** to the used oil tank **610**. The operator will then connect the quick disconnect **604** to piping **616** to connect the cart to the bulk tank **600**. Lastly, the operator will ensure that the three-way valve is positioned properly such that oil flow will move from piping **618** to piping **616**. These steps may also be conducted in different order. Lastly, however, the operator will energize the pump **606** using controls **626** and used oil will be drawn from the used oil tank **610**, through the inlet tube **622**, into nozzle **608**, and then into the pump **606** through piping **620**. From there, used oil will be forced through piping **618**, into the three-way valve **602**, piping **616**, and finally into the bulk tank **600**. When the used oil tank is emptied, or when the operator so desires, the flow may be stopped. The operator may then disconnect piping **616** and remove the inlet tube **622** from the port to safely move the cart throughout the facility.

[0119] FIG. 7B depicts an oil flow schematic of the used oil flow path in an exemplary cart while in a fryer service mode. Shown in FIG. 7B is the bulk tank **600** for the storage of used cooking oil. Also shown are the three-way valve **602** with quick disconnect **604**, pump **606**, nozzle **608**, and used oil tank **610**, all of which are associated with the cart **612**. Likewise, the fryer **614** is shown in a kitchen.

[0120] In order to transfer used cooking oil from the fryer **614** to the used oil tank **610** of the cart **612**, the inlet tube **622** of the nozzle **608** is placed within the used oil of the fryer **614**. The three-way valve **602** is also configured to permit flow from piping **618**, through the three-way valve, and into piping **628** which is configured between the three-way valve and the used oil tank **610**. When the pump **606** is energized via the control **626**, used oil flows from the fryer **614**, through the inlet tube **622**, into the nozzle **608** and piping **620** to the pump **606**, and then through piping **618**, three-way valve **602**, and piping **628**, finally arriving at the used oil tank **610**. When desired, the operator may de-energize the pump with the control **626**.

[0121] It will be appreciated that the flow paths of the fresh oil piping and the used oil piping do not interconnect. Therefore, at no point does used oil compromise the integrity of the fresh oil.

[0122] Given the foregoing four modes of operation, fresh oil can be transferred into the cart at the location of a bulk tank. The cart may then be moved to one or more remote fryers where the fresh oil may be transferred to the fryers utilizing the nozzle. In the meantime, used oil from the one or more fryers maybe drawn from the fryers into the cart. The cart can then be moved to the location of a used bulk oil tank for transfer of the used oil into the bulk tank. Using these methods of operations, facilities that could not be supplied with bulk oil may now be.

[0123] It will also be appreciated that the cart may be configured with components that do not include the nozzle described in U.S. Pat. No. 6,792,983. In such case, the cart may be configured as shown in FIGS. 6A-7B, with the exception that instead of a single nozzle there may be two nozzles, one in the system of FIGS. 6A-6B and a second in the system of FIGS. 7A-7B. In this way, fresh and used oil would still not mix. Of course, there can also be other piping configurations through which fresh and used oil does mix, for example by utilizing a single pump.

[0124] Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

1. A cooking oil system, said system comprising:
 - a first fresh oil tank capable of containing a volume of oil and being located at a first facility;
 - a second fresh oil tank located at a second facility, said second facility having at least two cooking areas with each of said cooking areas having at least one fryer, said second oil tank adapted to receive at least a portion of said volume of oil from said first fresh oil tank;
 - a third fresh oil tank associated with a first portable device at said second facility, said third fresh oil tank adapted to receive a second volume of oil, said second volume of oil being less than that of said at least a portion of said volume of oil from said first fresh oil tank, and said third fresh oil tank being adapted to receive oil from said second fresh oil tank.
2. The cooking oil system of claim 1, wherein receipt of the at least a portion of said volume of oil from said first fresh oil tank into said second fresh oil tank is via piping.
3. The cooking oil system of claim 1, wherein receipt of the at least a portion of said volume of oil from said first fresh oil tank into said second fresh oil tank is via an intermediary tank.
4. The cooking oil system of claim 1, wherein the at least a portion of said volume of oil from said first fresh oil tank is moved from said first facility to said second facility via an over the road vehicle.
5. The cooking oil system of claim 4, wherein said second fresh oil tank is associated with said over the road vehicle prior to being located at said second facility.

6. The cooking oil system of claim 1, wherein said second facility is one of a stadium, arena, cruise ship, casino, shopping mall, convention hall, or multi-story building.

7. The cooking oil system of claim 6, wherein said first portable device is sized and configured to fit within a service hallway of said second facility.

8. The cooking oil system of claim 7, further comprising a first waste oil tank at said second facility, said first waste oil tank being associated with a second portable device.

9. The cooking oil system of claim 8, further comprising a second waste oil tank located at said second facility.

10. A system for distributing fresh cooking oil, said system comprising:

- a first fresh oil tank located at a first facility, said first fresh oil tank capable of holding a first quantity of fresh cooking oil;

- an over the road vehicle adapted to transfer a second quantity of fresh cooking oil to a second facility, said second facility have at least two kitchens each with at least one cooking fryer;

- a second fresh oil tank, said second fresh oil tank adapted to receive at least some of said second quantity of fresh cooking oil;

- a first portable device located at said second facility, said first portable device adapted to receive a third quantity of fresh cooking oil and be moved to at least one of said kitchens.

11. The system of claim 10, wherein said second quantity of fresh cooking oil is less than said first quantity of fresh cooking oil.

12. The system of claim 11, wherein said third quantity of fresh cooking oil is less than said second quantity of fresh cooking oil.

13. The system of claim 10, wherein said second fresh oil tank contains said second quantity of fresh cooking oil when said over the road vehicle is transferring the second quantity to the second facility.

14. The system of claim 10, wherein said second fresh oil tank is located at said second facility while said over the road vehicle is transferring the second quantity of fresh cooking oil.

15. The system of claim 10, wherein said second facility is one of a stadium, arena, cruise ship, casino, shopping mall, convention hall, or multi-story building.

16. The cooking oil system of claim 15, wherein said first portable device is sized and configured to fit within a service hallway of said second facility.

17. The cooking oil system of claim 10, further comprising a first waste oil tank at said second facility, said first waste oil tank being associated with a second portable device.

18. The cooking oil system of claim 17, further comprising a second waste oil tank located at said second facility.

19. A system for distributing fresh cooking oil, said system comprising:

- a first fresh oil tank located at a first facility, said first fresh oil tank capable of holding a first quantity of fresh cooking oil;

- an over the road vehicle adapted to transfer a second quantity of fresh cooking oil to a second facility, said second facility have at least two kitchens each with at least one cooking fryer;

- a first portable oil receptacle located at said second facility, said first portable oil receptacle adapted to

receive a third quantity of fresh cooking oil and be moved to at least one of said kitchens.

20. The cooking oil system of claim **19**, wherein said first portable receptacle is a tank.

21. The cooking oil system of claim **20**, wherein said first portable receptacle is propelled via a powered propulsion system.

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